WORKED SOLUTION

STEP I 1992, Question 10

Mechanics • Recoil and Momentum

STEP I 1992, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q10

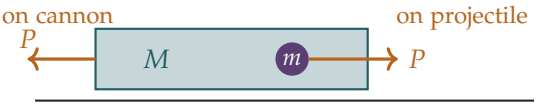
The key idea

The force P pushes the projectile forwards, but by Newton's third law the projectile pushes the cannon backwards with the same force P, and nothing horizontal stops the cannon recoiling. So the projectile does *not* travel a distance l relative to the ground while it is in the barrel: it travels l *relative to the cannon*. Both bodies have constant accelerations, so the cleanest route is to work with the relative motion.

Solution

The model

We treat the projectile and the cannon as particles moving along the same horizontal line, the floor as smooth (the cannon "slides freely"), and the force between projectile and barrel along the barrel as the constant P. Vertically, the weights are balanced by normal reactions (the floor on the cannon, the barrel on the projectile), so they play no part in the horizontal motion. Take the firing direction as positive.



The two accelerations

While the projectile is in the barrel, the only horizontal force on it is P forwards, and the only horizontal force on the cannon is the reaction P backwards (Newton's third law; the floor is smooth). By Newton's second law the accelerations are

$$a_{\text{proj}} = \frac{P}{m}, \qquad a_{\text{cannon}} = -\frac{P}{M}.$$

Both are constant, and both bodies start from rest.

Ask yourself Over what distance does the projectile accelerate: l, or something else?

The barrel has length l, but the barrel itself moves backwards. The projectile leaves when its displacement *relative to the cannon* reaches l. Relative to the cannon, the projectile starts from rest with constant acceleration

$$a_{\text{rel}} = \frac{P}{m} - \left(-\frac{P}{M}\right) = \frac{P}{m} + \frac{P}{M} = \frac{P(M+m)}{mM}.$$

The time in the barrel

Using $s = \frac{1}{2}at^2$ for the relative motion, the projectile emerges at the time T with

$$l = \frac{1}{2} \cdot \frac{P(M+m)}{mM}\, T^2, \qquad \text{so} \qquad T^2 = \frac{2mMl}{P(M+m)}.$$

The exit speed

Relative to the ground the projectile has constant acceleration P/m from rest, so at time T its speed v satisfies

$$\begin{aligned} v^2 = \left(\frac{P}{m}T\right)^2 &= \frac{P^2}{m^2} \cdot \frac{2mMl}{P(M+m)} \\ &= \frac{2PMl}{m(M+m)}. \end{aligned}$$

Since $v > 0$,

$$v = \sqrt{\frac{2PMl}{m(M+m)}},$$

which is the required result.

Another route: energy and momentum

The same answer comes from conservation laws. There is no external horizontal force, so horizontal momentum is conserved: if the cannon has speed V backwards at the moment of exit, then $mv = MV$. If the projectile moves x forwards and the cannon X backwards while the projectile is in the barrel, then $x + X = l$. The force on the projectile does work Px, and the reaction on the cannon, which points the same way as the cannon moves, does work PX; together they do work $P(x + X) = Pl$. Hence

$$Pl = \tfrac{1}{2}mv^2 + \tfrac{1}{2}MV^2 = \tfrac{1}{2}mv^2 + \tfrac{1}{2}M\left(\frac{mv}{M}\right)^2 = \tfrac{1}{2}mv^2\,\frac{M+m}{M},$$

which again gives $v^2 = \dfrac{2PMl}{m(M+m)}$.

Checks

- Heavy cannon: as $M \to \infty$ the factor $M/(M+m) \to 1$ and $v \to \sqrt{2Pl/m}$, the speed from a fixed cannon (work Pl all going to the projectile). A recoiling cannon always gives a slightly smaller speed, as it should, because some energy goes into the recoil.
- Units: Pl is energy and Pl/m has units of $\mathrm{m}^2\,\mathrm{s}^{-2}$, so v is a speed.
- Numbers: with $m = 0.1$, $M = 0.9$, $P = 9$ and $l = 0.5$ (SI units) the formula gives $v = 9\ \mathrm{m\,s^{-1}}$, with the cannon recoiling at $1\ \mathrm{m\,s^{-1}}$. Then momentum $0.1 \times 9 = 0.9 \times 1$ and kinetic energy $4.05 + 0.45 = 4.5 = Pl$. A step-by-step numerical simulation of both bodies, stopped when the separation reaches l, agrees with the formula for many random values of m, M, P and l.

Common slips

- Using $v^2 = 2(P/m)l$, which treats l as the projectile's distance relative to the ground and ignores the recoil.
- Forgetting Newton's third law and leaving the cannon at rest; the cannon slides freely, so it must recoil.
- In the energy method, taking the work done as P times the projectile's ground displacement only; the reaction on the cannon does work too.

Beyond STEP

This is a two-body problem, and the relative acceleration $P(M+m)/(mM)$ is P divided by the *reduced mass* $\mu = mM/(m+M)$. Any two bodies interacting only with each other behave, relative to one another, like a single body of mass μ; the same idea handles planetary orbits, binary stars and colliding particles, and it underlies rocket and recoil problems. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both the reduced mass and the momentum ideas here.

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